

Towards Algorithmic Reform: Low-Income Individuals Inclusion in AI/ML Literacy and Ethical Values-Informed Tool Design

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ABSTRACT

Poverty in the US is not invisible. A large number of Americans are low-income and experience homelessness. This population relies on scarce-resourced public services for survival and thriving. High-stake public service resource allocations are increasingly fueled by AI/ML to provide efficient and scalable services. While AI/ML tools are deployed with positive expectations, there is growing evidence of AI/ML causing invisible harm, often dismissed as inevitable. AI/ML tools are based on deficits and vulnerability ranking; they ignore the strengths of vulnerable individuals. This thesis aims to counteract existing exclusions and reduce AI/ML access barriers for low-income individuals, through their inclusion in AI/ML innovations and education, using mixed-methods experimental studies. Specifically, (1) designing and evaluating value-sensitive service-assessment tools that go beyond individual risk-factors and focus on strengths, (2) making AI/ML knowledge digestible for low-income individuals. Thereby, bringing about social change, AI/ML access, and algorithmic reform.

CCS CONCEPTS

• **Human-centered computing** → HCI theory, concepts and models; • **Social and professional topics** → User characteristics; • **Computing methodologies** → Machine learning algorithms.

KEYWORDS

Machine Learning (ML) Values, ML for Homelessness, Inclusion of impacted people in AI/ML, Low-income individuals, AI/ML literacy, Assets, Risk factors, Value-sensitive design, Inclusive design, Ethical values, ACM CHI, ACM CSCW

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1 BACKGROUND AND MOTIVATION

In the US, poverty is all too pervasive and persistent.¹ In 2022, more than 50 million Americans were classified as low-income and had an annual income of less than 125% of poverty.² Similarly, a large number of people experience homelessness every day; for example, 582,462 Americans are experiencing homelessness in 2023.³ Low-income individuals face barriers to supporting their daily needs, for example, access to healthcare, insurance, transportation, childcare support, access to education and stable income or job opportunities [1, 5, 15, 25, 49]. The survival of low-income people depends on public service, including welfare resources [17].

Artificial intelligence (AI), also known sometimes as powerful machine learning algorithms (ML), is now fueling public welfare services that affect poor and low-income individuals [8, 10, 15, 22, 43, 48]. For example, to offer efficient, low-cost, and scalable solutions to the large homeless population, there is a growing demand to automate public services by deploying AI/ML tools (e.g., VI-SPDAT assessment and ranking algorithms⁴) [27, 40, 44]. While housing prediction and classification tools support the staff members, they are sometimes detrimental to homeless people due to inaccurate, error-prone, or biased predictions. These tools are data-hungry and predict future behavior based on historical data, causing discrimination (e.g., replicating historical and pre-existing racial biases [20, 35]). AI/ML predictions divert the poor from deserved resources; they classify, criminalize, and punish the poor; they compromise their humanity, values, and self-determination abilities [17, 28].

Current public housing service assessment tools rely on vulnerability scores for outcome prediction and completely ignore the strengths and assets of vulnerable individuals [5, 21, 26, 53]. This deficit lens increases stigma and marginalization [54]. Most importantly, these people are not even aware that algorithms are being used to predict eligibility or service deservingness. Lack of transparency increases uncertainty, causing dissatisfaction and mistrust in public welfare services [14, 17, 35, 38, 50]. Homeless people are also almost always excluded from AI/ML technological innovations while directly impacted by them. Research shows that novelty and performance values are uplifted in ML homelessness service provision work [44]. Consequently, there is little or no consideration of human values in high-stake contexts (e.g., privacy and autonomy [19]). Building on these, the thesis will go beyond individual

¹<https://www.census.gov/newsroom/press-releases/2023/income-poverty-health-insurance-coverage.html>

²<https://justicegap.lsc.gov/resource/section-2-todays-low-income-america/>

³<https://todayshomeowner.com/general/guides/national-homeless-facts-and-statistics/>

⁴<https://everyonehome.org/wp-content/uploads/2016/02/VI-SPDAT-2.0-Single-Adults.pdf>

risk factors, focus on strengths, and sustainable assets, and center on ethical values for algorithmic tool design in the homelessness context.

The rapid proliferation of AI/ML caused an increasing effort toward making AI/ML knowledge more accessible for the general public [30, 34, 52]. However, the poor and the impacted people are mostly ignored in all these undertakings, despite being directly impacted by AI/ML. I argue that impacted low-income individuals should also have fair AI/ML learning opportunities, ensuring they are not left behind in the modern AI era. Thus, the second part of this thesis aims to support the AI/ML literacy of low-income individuals, enabling their meaningful participation in AI/ML discourse.

This dissertation will address the following Research Questions (RQs):

RQ1: What specific values are uplifted in ML or algorithmic approaches in homelessness service provision research?

RQ2: What specific assets are discovered in homelessness research?

RQ3: What are different stakeholders' perceptions of value-sensitive algorithmic tools in the homelessness context?

RQ4: How might we support low-income individuals' knowledge about algorithmic impacts and help them learn about supervised machine learning algorithms?

The first study (RQ1) was published in the ACM Conference on Human Factors in Computing Systems (CHI) 2023 [44]. The second study (RQ2) is currently under submission at the time of this paper was written. While the first two studies (RQ1-RQ2) are completed using a systematic literature review, the rest of the RQs will be investigated using mixed-method experimental studies. Background work (RQ1-RQ2) for the third study (RQ3) is currently completed, however, the design and evaluation part has not been completed yet. RQ4 is left intentionally broad. Data collection and analysis have already been performed for one of the research from the fourth RQ (RQ4), although the results are not reported in this paper. This work is expected to be submitted to the ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW) or related HCI conferences. In the following, I will briefly discuss RQ1-RQ2 findings and planned research for RQ3-RQ4.

2 CONTRIBUTIONS TO DATE

2.1 RQ1: Values Encoded in ML Homelessness Service Provision Research

Public housing is a scarce resource. Therefore, matching the large homeless population with scarce housing resources involves complex and sometimes difficult decision-making. Data science and ML approaches present valuable opportunities for homelessness service provision [11, 22]. Recent data science approaches developed algorithms for matching housing resources with high-risk clients [9], predict who will need assistance or cost more in the future, and forecast who will re-enter shelters [24]. All of these approaches rely on Homeless Management Information System (HMIS) data⁵, which is collected through collaborations and negotiations with agencies [9, 11, 47]. While the application of data science is capable of providing fast, low-cost, and scalable solutions for the staff, I would argue that homeless clients are vulnerable, and using their

sensitive data in the data science or ML pipeline and prioritizing different ML values (RQ1) can have unintentional harmful consequences and harm thereof (e.g., [41, 42]). To deeply assess this, this study performed a critical analysis of research papers identified through a systematic literature review to uncover (1) what values were uplifted in the research papers, and (2) the impacts of prioritization of those values. This work is published [44], and only partial findings are discussed below.

2.1.1 Methodology. This study followed a two-phase research method. First, the corpus was created using an in-depth systematic review of the literature of articles in CS/HCI and Science and Technology Studies (STS) that applied ML or statistical approaches in the context of homelessness [16, 40]. Second, thematic analysis was performed on the corpus [7]. A keyword-based search was carried out across the major databases, that is, the ACM digital library, Google Scholar, and IEEE Xplore, using various keywords, for example, "homeless", "homelessness", and "machine learning", including their combination. In total, we had 88 candidate papers from CS/HCI and STS literature, of which around 50 papers were quantitative. To focus on RQ1, 40 papers (16 CS/HCI papers and 24 STS-focused papers) were included in the thematic analysis. Both inductive and deductive thematic analysis was performed [7]. Deductive coding utilized 67 ML values proposed in Birhane et al. [6] GitHub page, to identify ML values that were uplifted in ML homelessness service provision research.⁶ Inductive coding was used to discover novel and surprising themes. Cross-coding or crosstabs was performed to discover the impact of dominant values on different values and themes.

2.1.2 Findings. Deductive analysis indicates that novelty, performance, and identifying limitations were the three most commonly occurring values in the corpus. The novelty value has often manifested in the choice of algorithms, dataset use, particular homeless population studied (e.g., homeless women, youth, veterans), novel predictor or risk factor identification, and experimental site selection (e.g., specific US state or city). CS-focused papers primarily focused on proposing novel ML algorithms for different prediction tasks, however, STS-focused papers focused on identifying novel risk factors. Novelty value was equally prioritized across CS (48%) and STS (52.5%) papers from a technical lens rather than a human-focused lens despite the sensitivity surrounding homelessness. Performance values such as accuracy were used in the traditional sense, i.e., for justifying numerical correctness. Crosstab analysis showed that 52.3% of the CS papers and 47.6% of the STS papers discussed performance. Although most papers included some discussion of the limitations of proposed ML interventions, however, those discussions often lacked human considerations in favor of technical limitations (e.g., dataset or sample limitations, limited predictor variable usage, and non-generalizability). The crosstabs further suggest that the STS (66.6%) papers tend to include study limitations more often than the CS (33.3%) papers.

The impact of these dominant values prioritization manifested in two ways: 1) unintended values collapse or values deprioritization⁷ and 2) human deprioritization. Inductive analysis showed

⁶<https://github.com/wagnew3/The-Values-Encoded-in-Machine-Learning-Research>

⁷I define "values collapse" as any ML values that were deprioritized or required re-definition as they break down or uphold their true definition as we know it in ML.

⁵<https://www.hudexchange.info/programs/hmis/>

that the meaning of several algorithmically charged values, such as privacy and reproducibility, was deprioritized, fragmented, or collapsed in this context. For example, privacy value collapsed a) when un-anonymized datasets (e.g., sensitive data such as medical and mental-health records, jail service use history, and SSNs) were shared with the research team, and b) when authors asked difficult or even invasive questions (e.g., non-prosocial survival practices, sexual behavior while on street) during interviews and survey data collection. Privacy was deprioritized through clients' lost autonomy about what information about them should or should not have been shared. Similarly, the reproducibility value collapsed a) when ML methods replicated vulnerability through algorithmic forecasting (e.g., someone who experienced homelessness during childhood or stayed in a homeless shelter or was born to a runaway parent is likely to experience homelessness during adulthood), b) when papers paid less attention to proposing transparent ML models and dataset use. In these ways, those research papers tend to reproduce vulnerability inadvertently and deprioritize those values.

The cross-coding assessment further suggests that prioritizing novelty and performance values was responsible for the ML values collapse mentioned above. Novelty and performance value prioritization also negatively impacted humans, causing human deprioritization through classification, categorization, prediction, and erasure through data pre-processing steps to algorithmic-specific parameter tuning steps in the data science pipeline [33]. Logistic regression was the most commonly used algorithm (20, 50%) in the corpus, and AUC/accuracy metrics were the most commonly used performance metrics (36, 90%), which contributed to ML values collapse. In RQ3, I will uplift the deprioritized values for designing a value-sensitive service assessment tool.

2.2 RQ2: Building Taxonomy and Critical Assessment of Homeless Person's Assets

In RQ2, the focus was on the positive aspects, strengths, and assets.⁸ Prior research has indicated that persons experiencing homelessness use different assets that help them thrive within the challenging homeless environment, assets help them in transition, and eventually move out of homelessness [21, 36, 45, 51]. Homeless people's assets are rarely showcased in HCI. Therefore, in RQ3, together with collaborators, I aspired to bring domain knowledge from homeless asset studies for HCI (1) to build a taxonomy of assets for future research and design, (2) to assess assets constraints and limitations that are recently discussed within HCI [53, 54]. This paper is currently under submission and only partial findings are reported below. The assets taxonomy developed from this work will be used in RQ3.

2.2.1 Methodology. First, the corpus was created using a systematic literature review of papers that reported homeless persons' assets; secondly, thematic analysis was performed to answer the RQs of this study [7]. The corpus was created using a keyword-based search using various terms such as, "homelessness", "assets",

"strengths", "resilience", "survival", "coping", and "asset-based approaches", including their combination. The final corpus included 40 papers, which were synthesized using thematic analysis. A bottom-up inductive thematic coding was applied, i.e., the analysis was driven by what is in the data. The research team systematically identified, and organized, insights into patterns of the dataset.

2.2.2 Findings. Our findings show that homeless persons used various external assets (17, 42.5%), strengths (22, 55%), resilience (14, 35%), and internal assets (5, 12.5%) to thrive on the streets. In the following, I will briefly discuss internal and external assets and introduce assets critical assessment.

Internal assets, also known as personal strengths, consist of the ability to exercise self-efficacy, empathy, problem-solving abilities, and self-awareness [21].⁹ We found that internal assets such as social competency and positive self-identity were associated with healthy development and resilience among homeless youth [26]. These assets also reduced their risk-taking behavior (e.g., violence, drug or alcohol use) among them. Bonds with pets and siblings were also identified as assets among youth [3]. For Aboriginal young women, reconnection with cultural knowledge and cultural efficacy, and restored identity helped them in the development of pride and hope, which in turn further contributed to their eventual transition out of homelessness [31, 32, 39]. In addition to internal assets, external assets (e.g., relationships, organizations) were also utilized among the homeless people [23]. For example, external assets available within the shelter, such as organization support and empowerment, contributed to a decrease in youth's distress during their shelter stay and improved their life satisfaction and internal assets. Young women who became pregnant or had babies served as a motivational factor (e.g., hope), and were an enabler of re-evaluation of personal choices and the safety in place [39].

Despite assets' positive impact on homeless persons' lives, we found several ways in which asset usage was nuanced and context-specific. For example, studies show that external peers help in decreasing youths' stress, by providing them with the necessary knowledge and learning opportunities for staying safe and strategies for surviving on the streets. However, we found contradictory evidence suggesting that these peer networks can also perpetuate and encourage youth to engage in extreme survival or non-prosocial strategies (e.g., selling drugs, prostitution) [18, 45]. Another subtle, yet problematic coping was observed when homeless persons' became content and found comfort within the current limited living arrangements. Consequently, they stop developing goals for the future or try to get housed. All this contradictory evidence shows the dual nature of assets, thus hinting at the complexity of designing technology with them [53, 54].

3 EXPECTED NEXT STEPS

The background work (RQ1-RQ2) of this proposal was based on a systematic literature review, the rest of the studies (RQ3-RQ4)

⁸Assets encompass all potential resources in the community, including financial resources, institutions, and property; and intangible and non-financial resources, consisting of people, connections, relationships, social competencies, positive values and life perspectives, individual knowledge, strengths, and capacities [5, 54] existing in the lives of persons experiencing homelessness.

⁹Internal assets generally identify characteristics and behaviors that reflect positive internal growth and development, e.g., these assets are about positive values and life perspectives, skills and adaptive characteristics, social competencies, positive self-identity, and commitment to learning [13, 21, 23]

¹⁰<https://outreach.msu.edu/documents/presentations/capcomm.pdf>

involve design and evaluation. Below, I will briefly discuss my proposed planned studies.

3.1 RQ3: Ethical Values-Informed Algorithmic Tool Design and Evaluation Study

The tool will be designed using the results of RQ1 and RQ2. First, findings from RQ1 indicate that current algorithmic or ML innovations in homeless service provision have been technical value-centered. These values prioritizations have caused important values to collapse and human deprioritization. Existing works also paid less attention to designing transparent algorithmic tools. Informed by these, I will uplift transparency and autonomy in RQ3. Second, current service prioritization tools exclusively rely upon causes of homelessness or risk-factors, and are deficit-focused. We need to uplift homeless persons' assets that enable them to thrive in harsh homeless circumstances, to counteract stereotyping and stigma. The taxonomy developed in RQ2 will be used for design in RQ3. Once the prototype is available, it will be evaluated with the relevant stakeholders, including impacted stakeholders to understand their perceptions. Following VSD, I identified both direct stakeholders: homeless frontline workers or staff, i.e., caseworkers and volunteers, and indirect stakeholders: persons experiencing or have experienced homelessness in the past 5 years, and low-income individuals with prior or no homelessness experience.¹¹ Prior studies have shown that low-income individuals are on the verge or at high risk of becoming homeless [1, 4, 5, 12, 29], therefore, including their perspective is equally important. One of the foreseeable challenges is engaging homeless staff and sheltered clients without a partnering organization. I have attended numerous meetings, reached out to organizations in Boston, and achieved some initial progress. In that vein, I look forward to gathering guidance and further directions at CHI'24 DC.

3.2 RQ4: AI/ML literacy studies

RQ4 aims to empower low-income individuals by providing them access to knowledge about algorithmic impacts and commonly used ML algorithms. AI's negative impact on poor and low-income individuals is prevalent (e.g., biases, and lack of transparency causing mistrust and uncertainty [17]), however, they are commonly excluded from AI/ML literacy interventions. Particularly, my thesis will center on those who are classified as low-income individuals because they access welfare resources, as well as public housing services (e.g., [4, 12, 29]).^{12, 13} These service allocation decisions involve evaluation using of vulnerability assessment or service prioritization tools, and thus low-income people need to be aware of AI/ML. RQ4 AI/ML literacy studies will employ a mixed-method experimental design and will be evaluated using crowd-sourcing platforms. Participants will be pre-screened using the monthly household income as a threshold based on the current federal poverty guidelines of the US [5, 25]. This study will challenge the normative

dominant narratives about who should or should not receive AI/ML literacy based on economic disadvantage and classification. The hope is to enhance the inclusion of impacted individuals in AI/ML. One limitation of this study is that it unintentionally prioritizes those who have access to technologies (e.g., internet connectivity, phone, laptop) and the required knowledge of how to use them.

All these works will culminate in contributing to HCI and ML scholarship as well as public services through *algorithmic reform* – by reducing AI/ML access and inclusion barriers. More precisely, the contributions include: 1) designing an ethical values-informed service assessment tool in the homelessness context and 2) disseminating AI/ML knowledge accessible to low-income individuals.

4 EXPECTED BENEFITS AND CONTRIBUTIONS AT CHI 2024 DOCTORAL CONSORTIUM

I am currently a third-year Ph.D. Candidate in the Khoury College of Computer Sciences at Northeastern University, supervised by Dr. Alexandra To. I have completed my Ph.D. comprehensive exam (i.e., Ph.D. proposal), and my projected dissertation completion date is between Late Fall 2024 to early Spring 2025. On average, Khoury Ph.D. students graduate in about six and a half years. Upon graduation, I aspire to pursue a research-oriented career either in academia or in the industry. I have never attended any Doctoral Consortium (DC). I have completed the preliminary works of my dissertation, however, I am still in the process of completing the rest of the planned studies (RQ3-RQ4) within the next year. ACM CHI DC 2024 will provide me with a great opportunity to gather new ideas and perspectives, a chance for critical reflection, and an opportunity to gain formative feedback on various aspects of my research due to the sensitivity that surrounds working with marginalized people. In addition, I look forward to meeting HCI and multi-disciplinary research experts, learning from them, connecting with other mentors and Ph.D. students, and identifying potential future collaboration opportunities.

There are several ways I can also contribute and be helpful to other ACM CHI DC'24 attendees. My current work entirely focuses on HCI, specifically, I draw inspiration from the value-sensitive design, human-centered AI [2, 19, 42, 44], and on social justice more broadly [14, 37, 46]. I have background experience in algorithm design and development, ML, mixed-methods study design, and advanced statistical data analysis. Also, I am experienced in paper publishing and have done peer-reviewing for quite some time. All these experiences make me equally beneficial to other attendees of the CHI DC. I can provide constructive and valuable feedback and help in interdisciplinary studies.

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¹¹“indirect stakeholders are those individuals who are also impacted by the system, though they never interact directly with it” [19]

¹²According to the 2023 U.S. federal poverty guidelines, individuals who had a household income below 150% of the poverty line, that is annual incomes below the range of 50,560 for a family of eight and 14,580 for an individual or family of size one, are classified as low-income.

¹³<https://aspe.hhs.gov/topics/poverty-economic-mobility/poverty-guidelines>

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